Mid Semester Examination-I (Sep 2020)
ITPE77
Data Warehousing and Data Mining

Time Allowed: 1 Hr
Max. Marks: 50 m

Note- **Attempt all questions.**

Q1 (a) Which steps are used in knowledge discovery process? Explain briefly.
(b) Neural networks are examples of classification or clustering?
(c) When a pattern is said to be interesting?
(d) Give an example of sequential pattern.

4, 2, 2, 2

Q2. (a) Define support and confidence in terms of probabilities and support counts.
(b) When association rules are called strong association rules?
(c) There are 12,000 (twelve thousands) transactions in a database. For the following association rule

Computer → UPS [Support 4%, Confidence 80%]

Compute the number of transactions that have entries at least of Computer.

4, 2, 4

Q3. (a) Let x, y and z are items belonging to a dataset S and * be a binary operator. When items x, y, and z are said to be closed under operator *.
(b) What is ant-monotone support property?
(c) If length of a frequent itemset is 10, compute
i. Total no. of frequent item sets
ii. Total no. of frequent item subsets
iii. Total no. of frequent 1-itemset
iv. Total no. of 5-itemsets

4, 2, 4